PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
FORM FOUR PRE - MOCK EXAMINATION
041 BASIC MATHEMATICS
(For Both School and Private Candidates)

TIME: 2:30Hours **March, 2025**

Instructions

1.This paper consists of sections A and B with fourteen questions.
2.Answer All Questions in both sections.
3.Each question in section A carries 06 marks while each question in section B carries 10 marks.
4.All necessary Working and Answers for each question must be shown clearly.
5.NECTA Mathematical tables and non-programmable calculators may be used.
6,All communication devices and any unauthorized materials are not allowed in the Examination room.
7.Write your examination number on every page of your answer booklet.

SECTION A (60 Marks)

01. a). work out (284 + 116) ÷ 40 + 5 x 3 – 8
    b). write 0.01498 to three significant figures
    c). simplify $\left(5\frac{1}{3} \times \frac{1}{4}\right) \div \left(1 \div ((3\frac{1}{4}) - \frac{5}{8})\right)$
02. a). solve for x in $3^{2x} = 3^{x+1}$ x 27
    b). if $\frac{logb}{log3} = \frac{log16}{log4}$ what is value of b?
03. a). given that x + y = 8 and $x^2 + y^2 = 40$. Find the value of xy.
    b). if μ = $\{a,b,c,d,e,f,g\}$, A = $\{e,f,g,\}$ and B = $\{d,e,f\}$. List all elements of.
        i. $n(A \cap B^C)^C$
        ii. $n((A \cup B) \cap (A^C \cup B^C))$ where c means complement or dash.
04. a). given that $\underline{a} = 3i + j$, $\underline{b} = 2i + 3j$ and $\underline{c} = i - 4j$. find the scalar value of p and q such that $p\underline{a} + q\underline{b} = \underline{c}$
    b). find the equation of the line whose slope is 3 and y- intercept is 4.
    c). use graphical method to solve $\begin{cases} x + y = 4 \\ 3x - y = 4 \end{cases}$
05. a). the areas of two similar polygons are $27cm^2$ and $48cm^2$. If the length of one side of the smaller polygon is 4.5cm. find the length of the corresponding sides of the larger polygon.
    b). the length of the diagonals of rhombus are 5cm and 12cm, find its area.
    c). twice the length of the rectangle is three times its width, if perimeter of this rectangle is 320. Find the width of the rectangle
06. a). convert 250 dollars to pounds if giving that 1 dollar is equivalent to 9900shillings and 1 pound is equivalent to 2200 shillings
    b). given that x is direct proportional to the square root of y and inversely proportional to z. given that x = 6 when y = 100 and z = 5. Find the value of z when x = 2 and y = 16.
07. a). a radio is bought for sh. 24,000. and sold for sh. 30,000. Find
        i. the profit made
        ii. the percentage profit
    b). find the simple interest on sh. 18,000 for 2 years at the rate of 6%

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08. a). if the 10th term of arithmetic progression is twice the 4th and the 6th term is 16. Find the sum of the 5th term and 6th term.

b). find the sum of the first four terms of geometric progression if the sum of the first two terms of a geometric progression eighteen where as the sum of the fourth and third term is one hundred sixty two.

09. June 16, 1985 sarwa started business with capital in cash 1,200,000/=
June 17, bought goods for cash 800,000/=
19, purchased shelves for cash 250,000/=
22, sold goods for cash 600,000/=
24, paid rent for cash 240,000/=
24, bought goods for cash 400,000/=
25, paid wages for cash 60,000/=
26, bought goods for cash 350,000/=
28, sold goods for cash 300,000/=
28, paid insurance for cash 100,000/=

a). enter the given transaction in cash account

b). extract a trial balance

10. a). factorize completely the following

i. $(x+1)^3 - x - 1$

ii. $3x^2 - 4$

b). if (x+2) and (x+3) are two factors of the quadratic equation $x^2 + px + q = 0$, find the value of p and q

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SECTION B. (40 marks)

11. a). the frequency distribution table below shows the number of pupils with their ages.

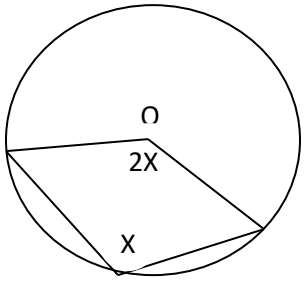
Ages (years)	3-6	7-10	11-14	15-18	19-22	23-26
frequency	32	30	19	11	06	3

i. find mean age (use assumed mean A = 12.5)

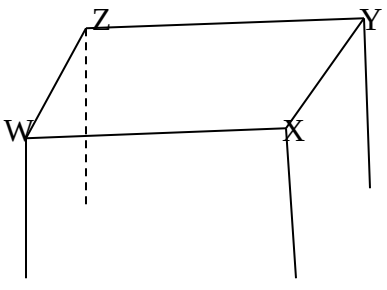
ii. find mode age

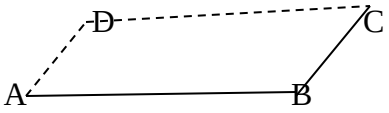
iii. estimate median age using cumulative frequency curve.

12. find the value of x. if O the center of the circle.



13. a). the figure below is the rectangular prism in which $\overline{AB}$ = 9cm, $\overline{BC}$ = 12cm, and $\overline{WA}$ = 3cm.





Calculate,

i. The length of AC
ii. the angle between WC and WA
iii. the volume of the Prism.

b). an airplane flies from Tabora (5^0S, 33^0E) to Tanga (5^0S, 30^0E) at 332km/h along a parallel of latitude. If nit leaves tabora at 3pm. Find the arrival time at tanga airport.

14. given that $f(x) = ax^2 + bx$, $f(1) = 5$ and $f(2) = 14$, find the value of x and y.

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